

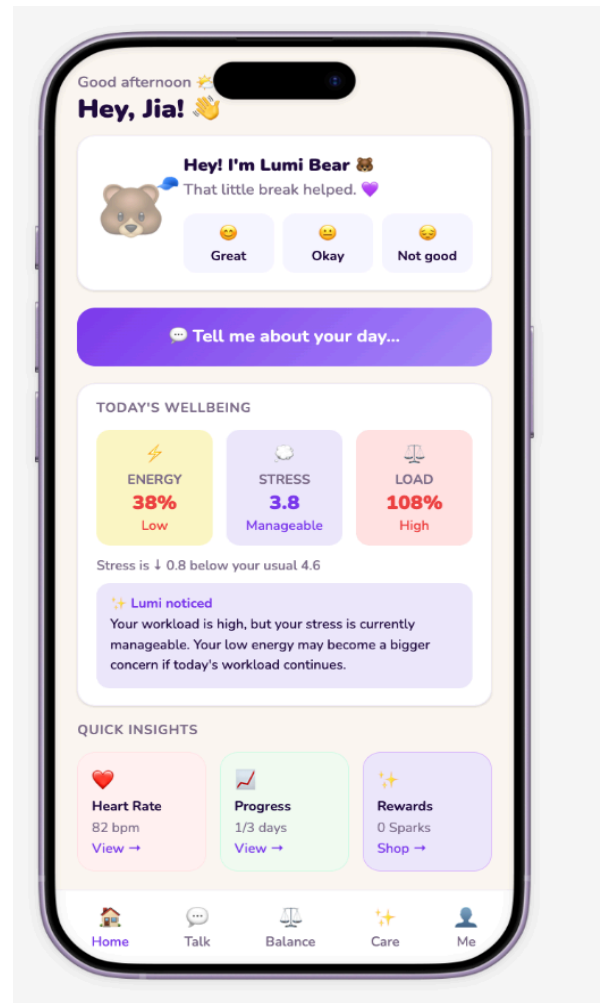
# LumiCare by Never404

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**Problem Statement:** Stress & Workload Manager

**Video Presentation:** [[Unlisted Youtube Link](#)]

**Presentation Slides:** [[Public Link](#)]



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# 1. Project Overview

## 1.1 The Problem.

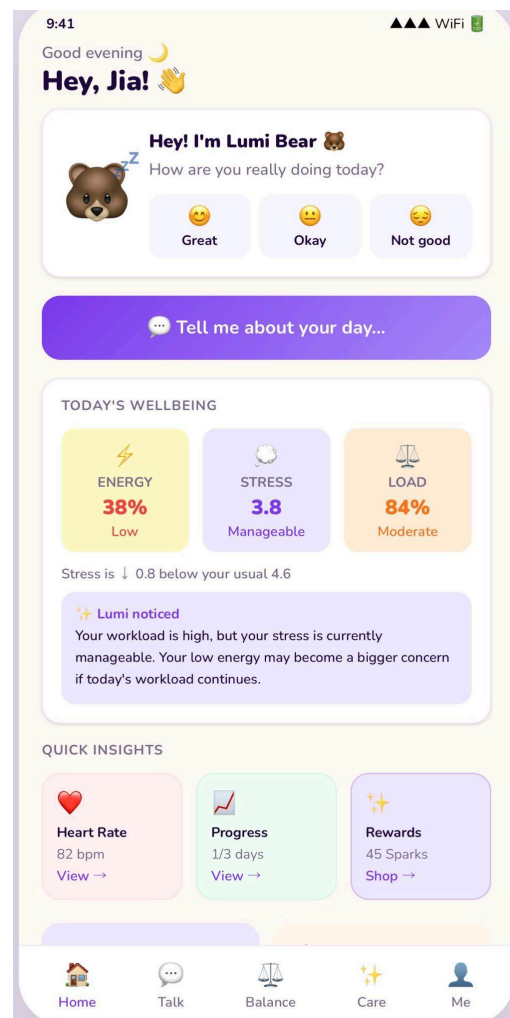
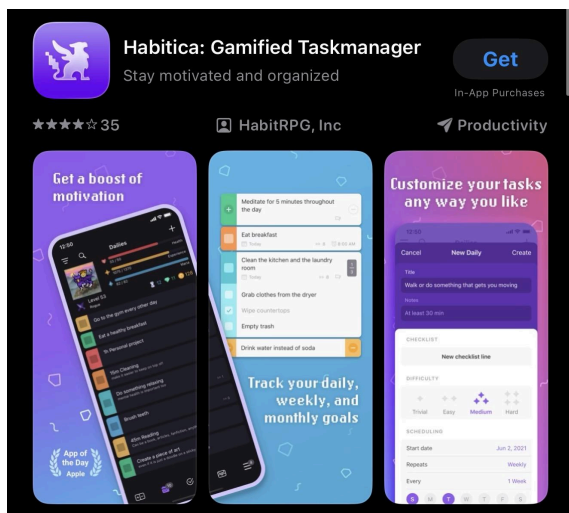
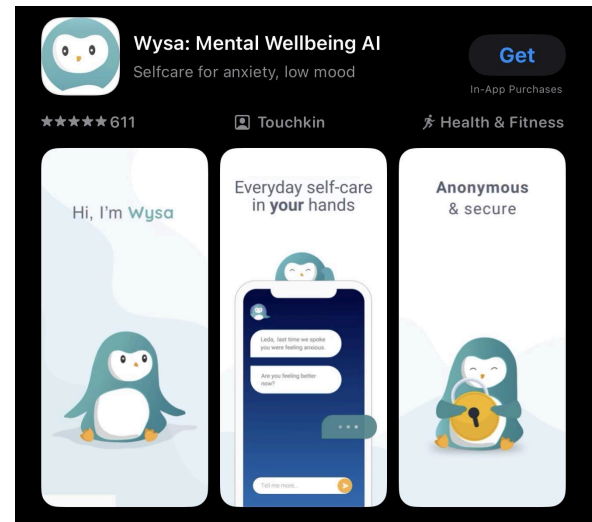
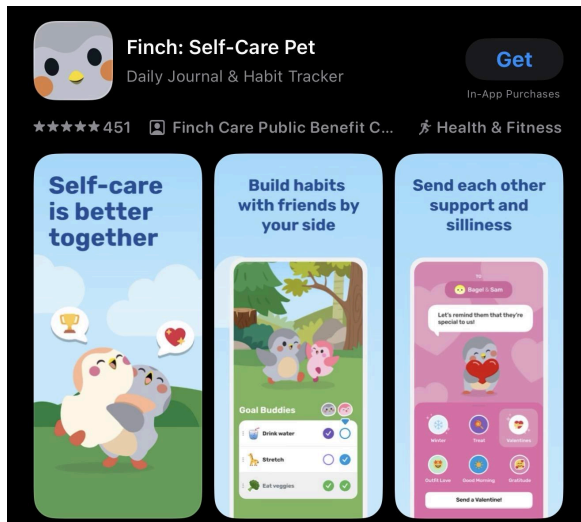
University students often have to balance multiple responsibilities at the same time, including assignments, classes, part-time work, social commitments, errands and personal wellbeing. The problem is not always that students have too much to do, but that the total weight of these responsibilities can build up gradually without them realising how overloaded they have become.

Students may also prioritise urgent tasks while postponing rest, social needs or personal recovery. As a result, they may continue pushing themselves even when their mental, physical or time capacity is already stretched.

The key stakeholders are **university students**, who are the primary users, as well as **universities and student support services**, which have an interest in students maintaining sustainable wellbeing and academic performance.

Existing solutions such as **productivity and task-management apps** help students organise what they need to complete, while **wellbeing or mental-health apps** focus more on mood, stress or relaxation. However, these approaches are often separated. A student may know their tasks and know how they feel, but still have no clear picture of **how everything together is affecting their overall capacity** or what they can realistically change.

| Existing App            | What It Does Well                                                                         | Why It Falls Short for Our Problem                                                                                                                                                                                  |
|-------------------------|-------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Finch</b>            | Uses a virtual pet to encourage self-care, mood tracking, reflection and daily goals.     | Focuses mainly on <b>general self-care and habit building</b> . It does not specifically analyse a university student's overall workload across different areas such as mental, time, physical, social and errands. |
| <b>Habitica</b>         | Turns tasks and habits into a game using rewards, levels, quests and an avatar.           | Mainly focuses on <b>productivity and completing tasks</b> . It does not consider the user's emotional state or whether their workload is becoming overwhelming.                                                    |
| <b>Wysa</b>             | Provides an <b>AI conversational companion</b> for emotional wellbeing and self-help.     | Focuses mainly on <b>emotional support</b> and does not combine this with task or workload management, gamified rewards or a virtual pet.                                                                           |
| <b>Our Proposed App</b> | Combines an <b>AI companion, virtual pet, workload management and wellbeing support</b> . | Designed specifically to help <b>university students understand, manage and rebalance their overall workload</b> rather than focusing on only one aspect.                                                           |



## 1.2 Our Solution.

**Lumi is an AI-powered companion designed to help university students understand and manage their overall workload and wellbeing.**

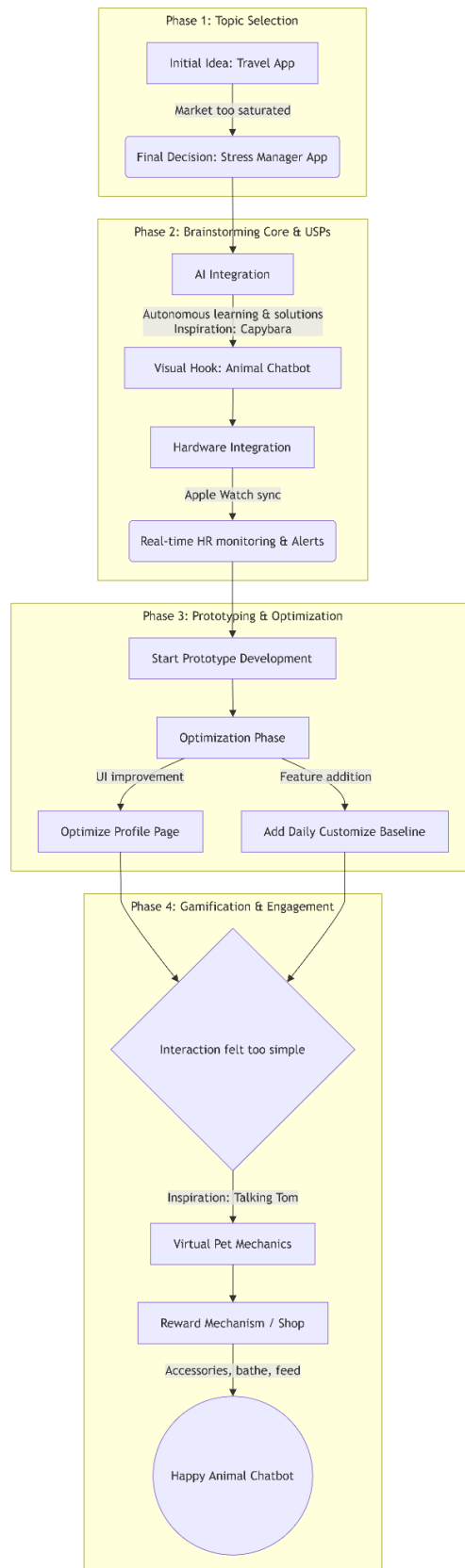
- **AI Companion Chat** — Talk naturally about your day, tasks, commitments and emotions.
- **Context Understanding** — Extract relevant workload and wellbeing information from conversations.
- **Personalised Load Overview** — Visualise workload across mental, time, physical, social and everyday-life areas.
- **Heart Rate Insights** — Use optional smartwatch/heart-rate data as an additional wellbeing signal.
- **Pressure Point Analysis** — Identify which responsibilities are contributing most to overload.<sup>5.5</sup>
- **Interactive Rebalancing** — Move or adjust commitments and immediately see the projected impact on overall load.
- **Recovery Mode** — Encourage small recovery actions such as nature walks, deep breathing, me time and talking to someone.
- **Virtual Companion** — The companion reflects the user's state, becoming tired when the user is overloaded and recovering as the user takes care of themselves.
- **Sparks & Rewards** — Earn Sparks through self-care and use them to customise the companion.
- **3-Day Recovery Challenge** — Build consistent recovery habits and recharge the companion.
- **Reflection & Progress** — Help users recognise patterns in their workload and recovery over time.

## 2. Ideation & Process

### 2.1 Ideas We Considered

| <b>Idea</b>                                         | <b>Why kept / dropped</b>                                                                 |
|-----------------------------------------------------|-------------------------------------------------------------------------------------------|
| <b>AI Companion + Virtual Pet (Chosen)</b>          | Combines emotional support with workload management in an engaging way.                   |
| <b>Workload Visualiser / Load Meter (Chosen)</b>    | Helps students see how their responsibilities add up to overall overload.                 |
| <b>AI Conversation → Workload Analysis (Chosen)</b> | Let students share naturally without manually categorising everything.                    |
| <b>Interactive Load Rebalancing (Chosen)</b>        | Allows students to adjust commitments and see how their workload changes.                 |
| <b>Recovery System + Sparks (Chosen)</b>            | Makes self-care an active part of the experience through recovery activities and rewards. |
| <b>Heart Rate / Smartwatch Signal (Chosen)</b>      | Adds another wellbeing signal; simulated for the MVP.                                     |
| <b>3-Day Recovery Challenge (Chosen)</b>            | Encourages consistent recovery instead of one-time actions.                               |
| <b>Pure Productivity / To-Do App</b>                | Dropped — Too focused on task completion.                                                 |
| <b>Standalone Mood / Stress Tracker</b>             | Dropped — Does not connect feelings with workload.                                        |
| <b>Generic AI Mental-Health Chatbot</b>             | Dropped — Our focus is everyday student workload and recovery, not therapy.               |
| <b>What-If Planning Tool</b>                        | Dropped — Rebalancing already provides the useful scenario-testing function.              |

## 2.2 Ideation Boards

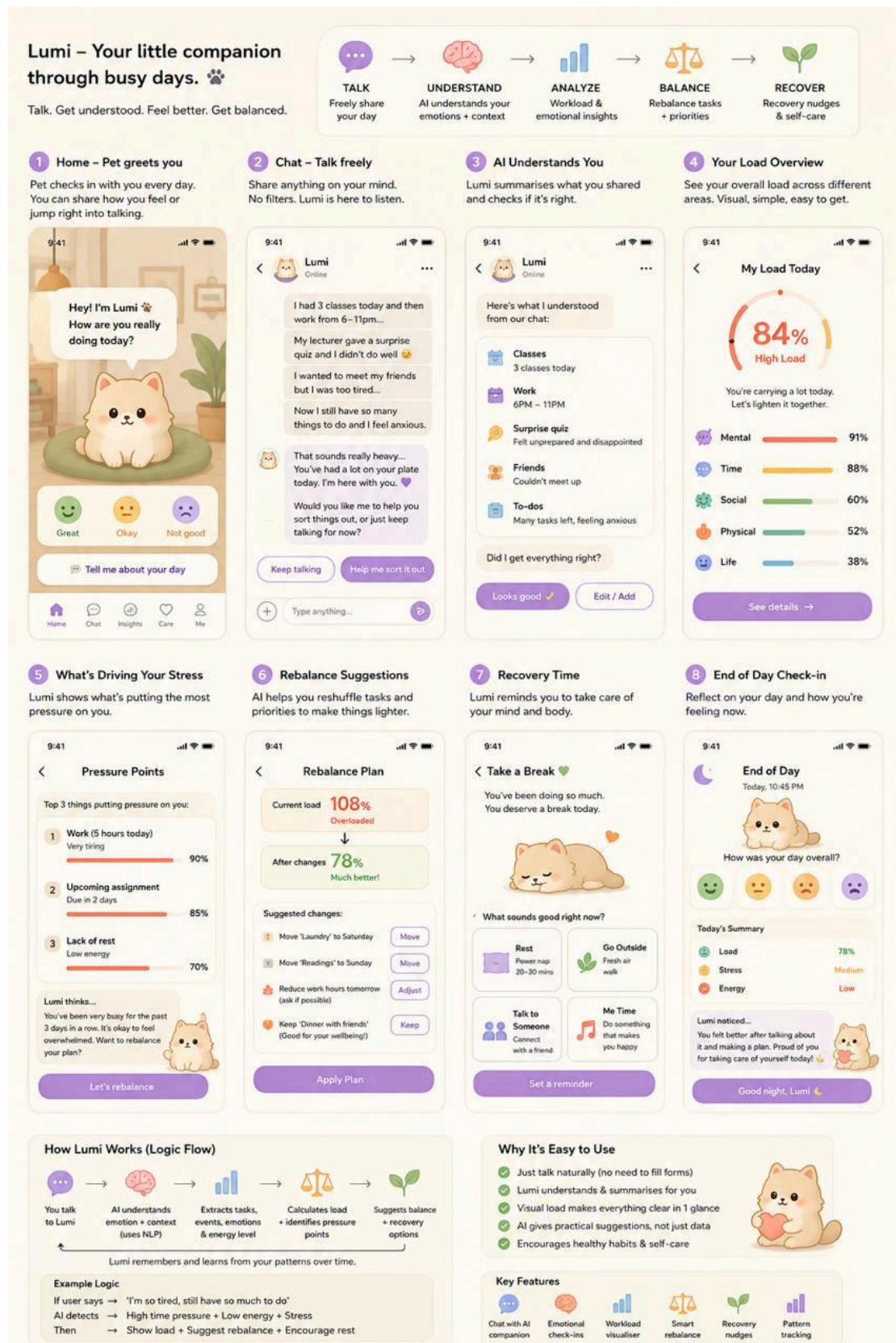


## 2.3 Mentor Consultation

| Date     | Mentor      | Feedback Received                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | What Was Changed                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6/9/2026 | Teh Ming En | <ul style="list-style-type: none"><li>- Home page too complex</li></ul> <p>The overall idea and design were good and well presented.</p> <ul style="list-style-type: none"><li>- Since the application relies heavily on AI, the team should consider the API requirements and budget.</li><li>- The UI was sweet and visually appealing, but it was slightly too colourful.</li><li>- The team could consider customising the 3D mascot further.</li><li>- The homepage contained too much information and should be simplified.</li></ul> | <ul style="list-style-type: none"><li>- Make home page to more simple and clear</li><li>- The team will review the AI API usage and budget during the development phase.</li><li>- The current colour scheme was retained as it fits the friendly and warm concept of the application.</li><li>- Further customisation of the 3D mascot will be considered during development.</li><li>- The homepage was simplified by removing unnecessary information and keeping only the main elements, resulting in a cleaner and clearer design.</li></ul> |

### 3. Design & Prototype

UI Prototype: [  Figma ]



## 4. What Makes It Different

LumiCare differentiates itself from existing productivity, wellbeing, and virtual companion applications by integrating **AI-powered conversation, workload analysis, workload rebalancing, recovery support, and gamification** into a single system. Rather than addressing productivity and wellbeing as separate areas, LumiCare is designed to help university students understand how their academic, work, social, personal, and emotional responsibilities collectively affect their overall capacity.

The originality of Lumi lies primarily in the **integration and interaction between these features**, rather than in treating each individual feature as completely new. The proposed system creates a continuous process of **understanding, analysing, rebalancing, and recovering**, specifically targeted at the everyday workload challenges faced by university students.

### 4.1 Novel Features

#### 1. AI Conversation-Based Workload Analysis

LumiCare allows users to describe their day, responsibilities, and feelings through natural conversation rather than requiring them to manually enter and categorise every task. The AI analyses the conversation and extracts relevant information such as classes, work commitments, assignments, social activities, emotions, and other sources of pressure.

The novel aspect is the use of **natural conversation as the primary input for workload analysis**, reducing the effort required from users while connecting emotional context with practical responsibilities.

#### 2. Multi-Dimensional Workload Visualisation

Instead of representing workload through only the number of tasks completed or a single stress score, LumiCare evaluates workload across multiple dimensions, including **mental, time, physical, social, and everyday-life factors**. This provides users with a more comprehensive view of their overall situation and helps them identify areas that may be contributing to overload.

### 3. Pressure Point Identification

LumiCare analyses the user's workload to identify the responsibilities that contribute most significantly to their current pressure level.

This provides a distinction from conventional task-management systems, as LumiCare focuses not only on what needs to be completed, but also on which responsibilities are contributing to the user's overall pressure.

### 4. Interactive Workload Rebalancing

A key feature of LumiCare is its interactive rebalancing mechanism. After identifying the user's pressure points, LumiCare can suggest adjustments to their commitments. Users can move or modify selected responsibilities and observe the projected change in their overall workload.

This introduces a decision-support and scenario-testing approach to workload management. Instead of simply providing a list of tasks, LumiCare allows users to explore possible changes and understand their potential impact before making decisions.

### 5. Recovery Integrated into Workload Management

LumiCare treats recovery as an integral component of workload management rather than as a separate wellbeing activity. When the system identifies that a user is experiencing a high workload, it can recommend appropriate recovery activities such as taking a break, going outside, spending time with someone, or having personal time.

The key distinction is that **rest and recovery are considered part of achieving a sustainable workload**, rather than being treated as optional activities after all tasks have been completed.

## 6. Adaptive Virtual Companion

Lumi incorporates a virtual companion that reflects the user's workload and recovery progress. The companion can become tired when the user experiences a high workload and recover as the user engages in recovery activities.

Unlike a conventional decorative mascot, the companion is connected to the user's interaction with the system. This provides an emotional and visual representation of the user's wellbeing state.

## 7. Recovery-Based Gamification

Lumi introduces a **Sparks reward system** that encourages users to participate in recovery activities. Users can earn Sparks through healthy recovery actions and use them to customise their virtual companion.

This provides a different approach to gamification. Instead of primarily rewarding users for completing more tasks, Lumi uses rewards to encourage users to **maintain a balance between productivity and recovery**.

## 8. Short-Term Recovery Challenge

The 3-Day Recovery Challenge encourages users to complete recovery activities consistently over several days. The purpose is to encourage repeated recovery behaviour rather than relying on one-time relaxation activities.

The feature also connects recovery behaviour with the virtual companion, allowing users to observe a sense of progression as they maintain their recovery activities.

## 9. Optional Heart-Rate Integration

Lumi incorporates heart-rate information as an additional indicator of the user's wellbeing. In the MVP, simulated heart-rate data is used to demonstrate how physiological information could complement the workload and emotional information obtained through conversation.

This creates the possibility of combining **self-reported information with physiological signals** in future versions of the system.

## 10. Long-Term Reflection and Pattern Recognition

Lumi records workload, recovery, and reflection information to help users recognise patterns over time. Instead of focusing only on the user's current tasks or mood, the system can provide a longer-term perspective on how workload and recovery behaviours change.

This supports the broader goal of helping students develop greater awareness of their workload and recovery habits.

### 4.2 Comparison with Existing Solutions

LumiCare's differences from existing solutions can be summarised in the following comparison table.

| Application | Primary Focus                     | Limitations for the Identified Problem                                                                                          | LumiCare's Differentiation                                                                                       |
|-------------|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Finch       | Self-care, habits and virtual pet | Focuses mainly on self-care and habit-building without analysing the user's overall workload across different areas.            | LumiCare combines a virtual companion with workload analysis, pressure identification, rebalancing and recovery. |
| Habitica    | Productivity and gamification     | Primarily encourages users to complete tasks and habits without considering their emotional state or overall workload capacity. | LumiCare considers both productivity and wellbeing, while rewarding recovery rather than only task completion    |
| Wysa        | AI emotional wellbeing support    | Primarily focuses on emotional support and self-help rather than                                                                | LumiCare uses AI conversation to connect emotional context with                                                  |

|          |                                   |                                |                                                                                                                                                                     |
|----------|-----------------------------------|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                   | practical workload management. | workload analysis and practical rebalancing.                                                                                                                        |
| LumiCare | Workload and wellbeing management | -                              | Integrates AI conversation, multi-dimensional workload analysis, pressure identification, interactive rebalancing, recovery, gamification, and a virtual companion. |

The comparison builds upon the existing-solution analysis in Section 1, where Finch, Habitica, and Wysa were identified as solutions that address individual aspects of self-care, productivity, or emotional wellbeing but do not combine these elements around university students' overall workload.

The main innovation of LumiCare is the connection between workload management and wellbeing within a single user flow. Existing solutions may help users complete tasks, track habits, monitor emotions, or practise self-care. LumiCare combines these functions so that information obtained from one stage can influence the next stage.

The proposed workflow begins with Natural Conversation, where users communicate their workload, feelings, and daily challenges in a simple and conversational manner. The system then applies AI Context Understanding to interpret the user's input and identify relevant tasks, emotions, deadlines, and contextual information. This information is processed through Multi-Dimensional Workload Analysis, which evaluates workload across different dimensions such as task volume, urgency, time pressure, and emotional strain. Based on this analysis, the system performs Pressure Point Identification to detect the main sources of stress or potential overload. Users can then participate in Interactive Workload Rebalancing, where the system suggests adjustments such as prioritising tasks, rescheduling activities, or

redistributing workload. Afterward, Recovery Recommendations are provided based on the user's current workload and stress level, including suitable breaks, recovery activities, or coping strategies. Finally, Reflection and Progress Tracking allows users to review their workload patterns, monitor changes over time, and gain insights into their progress and overall well-being.

This approach changes the purpose of a workload-management application from simply helping users complete more tasks to helping them manage their responsibilities in a more sustainable way.

Therefore, LumiCare's competitive advantage is not based on a single isolated feature. Its differentiation comes from the integrated ecosystem of AI understanding, workload visualisation, interactive rebalancing, recovery support, and an adaptive virtual companion, designed specifically for the workload and wellbeing challenges experienced by university students. These elements were selected during the team's ideation process as core components of the proposed solution.

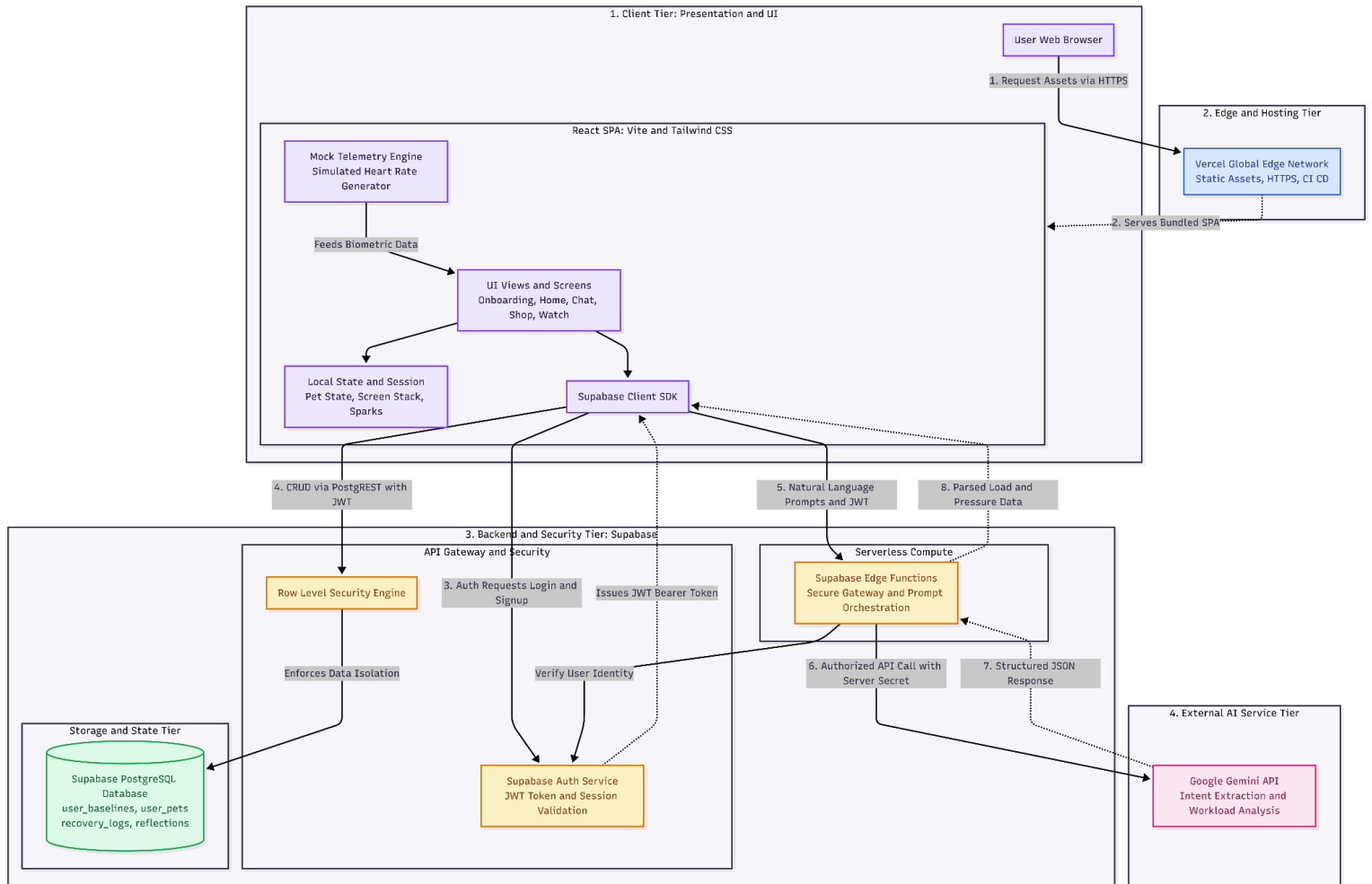
## 5. Technical Architecture & Feasibility

### 5.1 Tech stack

| Component      | Technology              | Purpose                                                                                                             | Reason we choose it                                                                              | Expected Constraints                                                                      |
|----------------|-------------------------|---------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| Prototype      | Figma                   | Design the interface and interactive prototype                                                                      | Easy to collaborate on and allows the team to quickly test and refine the Lumi user flow         | Figma is only a design/prototyping tool and cannot serve as the final functional frontend |
| Frontend       | React                   | Implement the actual Lumi web interface based on the Figma prototype                                                | Component-based, reusable, and suitable for building an interactive web application              | The team will need to convert the Figma prototype into functional React components        |
| Backend        | Supabase Edge Functions | Handle server-side logic and communication with external APIs                                                       | Reduces the need to maintain a separate backend server and provides a simple serverless approach | Free-tier usage limits and execution limits may apply                                     |
| Database       | Supabase PostgreSQL     | Store user profiles, baseline information, tasks, workload, recovery activities, mood, rewards, and reflection data | Integrated with Supabase backend services and suitable for structured relational data            | Free-tier database storage and usage are limited                                          |
| Authentication | Supabase Auth           | Handle user registration and login                                                                                  | Already integrated with Supabase and reduces the need to build authentication from scratch       | Free-tier limits and authentication configuration need to be considered                   |

|                 |                                 |                                                                                                                                                             |                                                                                                                |                                                                               |
|-----------------|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| AI API          | Google Gemini API               | Understand natural language conversations, extract workload information, identify pressure points, generate rebalancing suggestions and reflection insights | Provides an accessible AI API with a free development tier and supports the main AI functionality of Lumi      | Free-tier rate limits, API latency, and possible inaccurate AI interpretation |
| Smartwatch Data | Simulated Heart-rate Data (MVP) | Demonstrate the heart-rate feature                                                                                                                          | Allows the team to demonstrate the concept without the complexity of integrating multiple smartwatch platforms | Real smartwatch integration will not be included in the initial MVP           |
| Hosting         | Vercel                          | Host and deploy the functional web application                                                                                                              | Simple deployment workflow and suitable for React/Next.js projects with a free tier                            | Free-tier resource and usage limits may apply                                 |

## 5.2 System architecture diagram



## 5.3 Build plan & scope

### Sprint 1:Data Foundation & Intake flow

- Configure Supabase project,establish constrains and enforce basic **Row-Level Security policies**
- Port Figma designs into modular React components using Tailwind CSS and Vite.
- Implement the multi-step **OnboardingScreen and BaselineScreen**,persisting answers directly to the user's Supabase profile.

### Sprint 2:Secure AI Pipeline & Dynamic Load Engine

- Deploy a Supabase Edge Function as the secure server-side layer for AI requests,validating authorization headers and **preventing exposure of Google Gemini API key to the client**
- Connect the conversational flow from ChatScreen=>UnderstandingScreen=>LoadScreen=>PressureScreen using the structured AI response
- Implement the stateful RebalanceScreen, allowing users to toggle commitments while dynamically recalculating and animating their projected workload reduction.

### Sprint 3:Recovery Economy & Simulated Telemetry

- Implement RecoveryScreen with recovery-action logging.Completing an action will atomically update the user's sparks balance and append a corresponding record to the database
- Build RewardsScreen to manage reward unlocking and dynamically attach unlocked companion SVG assets.
- Implement the three-day recovery streak mechanism in ChallengeScreen, triggering the recharged state and reward drop after successful completion.
- Develop a time-series mock heart-rate generator to provide WatchScreen with realistic simulated afternoon stress-spike patterns.

#### Sprint 4: Integration, Evaluation & Deployment

- Configure continuous deployment from **GitHub** => **Vercel**
- Implement defensive error handling, including AI rate-limit fallbacks, request timeouts, and graceful offline/loading states
- Conduct functional and responsive testing across mobile viewport configurations, including Safari and Chrome emulation
- Prepare technical documentation and evaluation metrics, including AI response latency, system reliability, and user task-completion rates